Nucleo de AeroEspacial

ASTRO

Electronics Report

Author:

Lorem Ipsum (31416) lorem.ipsum@campus.fct.unl.pt
Lorem Ipsum (31416) lorem.ipsum@campus.fct.unl.pt

2025/2026 – 1st Semester, DEEC

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1 Introduction

This document presents the Design Review of the avionics and simulation systems developed for the sounding rocket project by the Astro team. The objective of this review is to describe the design choices, implementation details, and validation methods adopted for the onboard electronics, communication systems, ground station, and trajectory simulations, while demonstrating compliance with mission and competition requirements.

The avionics subsystem is responsible for sensor data acquisition, flight state estimation, telemetry transmission, recovery system activation, and overall mission safety. Reliability, redundancy, and robustness under high-dynamic flight conditions were key drivers throughout the design process. Component selection and system architecture were therefore guided by prior flight experience, established aerospace practices, and simulation-based validation.

In parallel, detailed trajectory simulations were conducted to predict flight performance, assess aerodynamic behaviour, and support design decisions related to propulsion, stability, and recovery systems. These simulations serve both as a verification tool during development and as a means of reducing risk ahead of flight operations.

This report is structured as follows: Section 2 describes the sensors and analog circuitry; Section 3 presents the selected microcontroller; Section 4 details the communication system; Section 5 outlines the ground station architecture; and Section 6 discusses the simulation framework and validation methodology. Supporting material is provided in the annexes.

2 Electronic Components

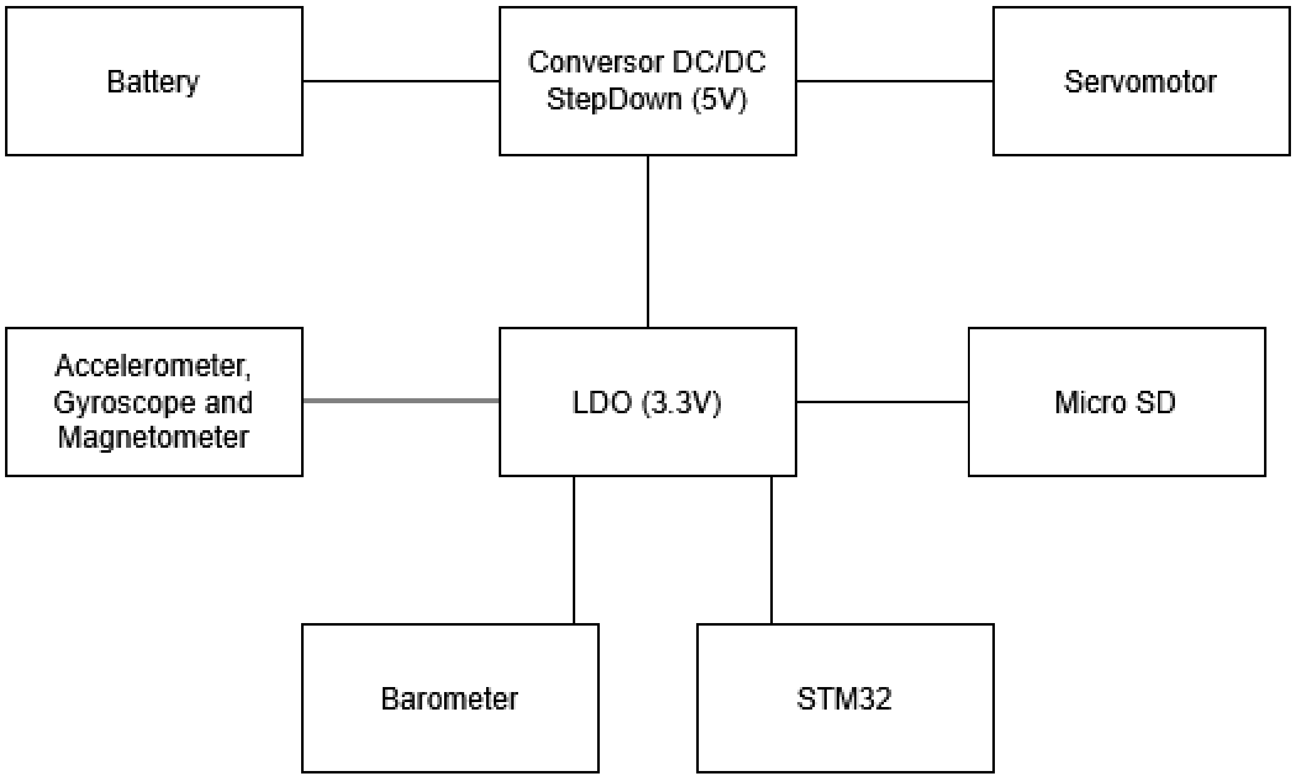

Figure 1: Power Supply

3 SRAD Flight Computer Software Architecture

Table 1: SRAD avionics electronic components quick specifications

Device	Supply (V)	Current	Power (mW)	Interface	Role	Notes
STM32F411 Microcontroller	1.8–3.6	–	–	I2C / SPI / UART	Main flight computer	Cortex-M4 MCU
ICM-45686 6DOF IMU	3.3	34 mA	112	I2C / SPI	Inertial sensing	High-rate motion tracking
ADXL375 3DOF Accelerometer	2.0–3.6	145 µA	0.5	SPI / I2C	High-G acceleration	Up to ±200 g
LIS2MDL Magnetometer	1.71–3.6	200 µA	0.66	I2C	Heading reference	Low-power compass
MS5607 Barometer	1.8–3.6	1.4 mA	4.6	I2C / SPI	Altitude estimation	High-resolution pressure sensor
NEO-M9N GPS Module	3.6	–	–	UART	Positioning	Multi-constellation GNSS
SX1278 LoRa (Ra-02)	1.8–3.3	105 mA	346	SPI	Telemetry link	Long-range RF comms
SparkFun microSD Module	3.3	20 mA	66	SPI	Data logging	Flash storage interface

Table 2: COTS avionics system specifications

Device	Supply (V)	Current	Power (mW)	Interface	Role	Notes
CATS Vega Flight Computer	7–24	100 mA	321.75	UART / CAN (var.)	Recovery control system	Integrated flight computer + deployment logic

Table 3: Bill of Materials

Item	**Status**	**Brand**	**Supplier**	**Qty**	**Unit**	**Total**
STM32F411 Development Board	In Lab	STMicroelectronics	Fruugo	1	8,95	8,95
Cats Vega	Need to order	catsystems	catsystems.io	2	395,73	791,46
GNSS M6N	In Lab	u-blox	Donated	2	0	0
LSM6DSMTR (Accel + Gyro)	Need to order	STMicroelectronics	Farnell / DigiKey	1	3,07	3,07
MS5607-02BA Barometer	Need to order	Measurement Specialties	DigiKey	1	2,8	2,8
LIS3MDL Magnetometer	Need to order	STMicroelectronics	PTRobotics	1	9,16	9,16
LiPo Battery 2200mAh 7.4V 2S1P	Need to order	Gens Ace	Techinn	2	17,95	35,9
LT317A PMU	Shipping	–	Mauser PT	2	0	0
Capacitor 1uF	In Lab	–	Mauser PT	2	0	0
Capacitor 0.1uF	In Lab	–	Mauser PT	2	0	0
FQP30N06L MOSFET	Shipping	–	Mixtronica	4	0	0
Antennas	Manufacturing in house	Nuar-Astro	–	2	20	40
LoRa SX1278 RA-02	Shipping	Ai-Thinker	LojaPM	2	9,89	19,78
					Grand Total	**911,12**